v1.0Axion Deep Dive Emulation

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Axion Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** `v1.0Axion Deep Dive Emulation V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Axion search method	Microwave cavity haloscope general description	HAYSTAC Phase II (2023): quantum squeezed states achieve 15.1 dB below SQL — doubles scan rate	MEDIUM	Search technique evolution changes simulation sensitivity parameter range
BEC phase transition	Conceptual description	V2.0: BEC coherence length xi = hbar/(m_a*v_rms). Phase transition implemented — switch from particle to coherent wave mode above critical density	HIGH	Missing BEC threshold is major omission for FDM cosmological simulation
Fuzzy DM mass constraint	m_a range from pre-2022 data	JWST 2023: m_a > 10^-21 eV from galaxy morphology — constrains simulation parameter space	HIGH	V1.0 allows simulation parameter values JWST rules out

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
CASPEr-electric	Not covered	Below 10^-8 eV axion search — first results 2024 from NMR-like axion-nucleon coupling	LOW	New detection method to model in simulation detector objects
ORGANIZE frequency range	Not covered	ORGAN 2023 at 26 GHz: higher-mass axion (m_a ~ 110 μeV) parameter space opened	LOW	Extends valid axion mass range for simulation

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** BEC, Fuzzy
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Axion Deep Dive Emulation V1.0 archived | V2.0 is the active specification as of June 2026